EECS 484: Database Management Systems

Winter 2026

Instructor: Barzan Mozafari

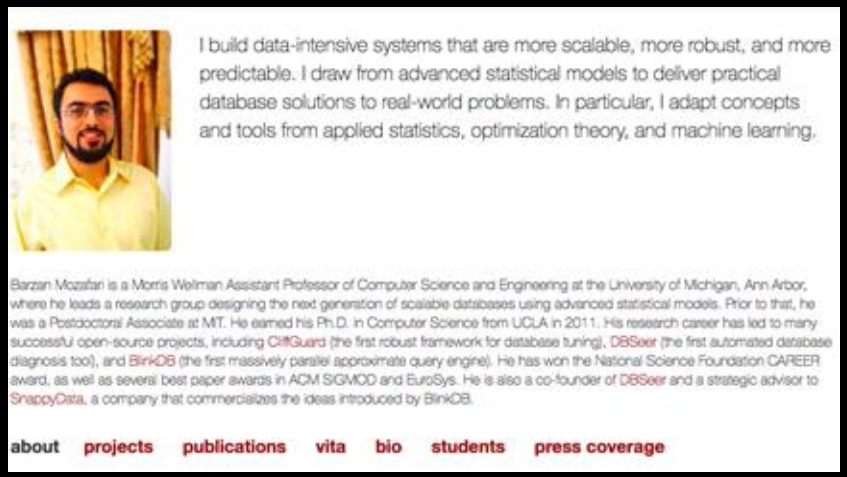
University of Michigan, Ann Arbor

Today's Goals

1. Getting to know your instructor
2. Why study databases
3. Logistics for the rest of the semester

About Your Instructor

- Short Bio
  - BS (SBU)
  - MS, PhD (UCLA)
  - PostDoc (MIT)
- Research Interest
  - Building large-scale data-intensive systems
- Passions
  - Tea
  - Tea
  - Tea
  - Databases + AI



I build data-intensive systems that are more scalable, more robust, and more predictable. I draw from advanced statistical models to deliver practical database solutions to real-world problems. In particular, I adapt concepts and tools from applied statistics, optimization theory, and machine learning.

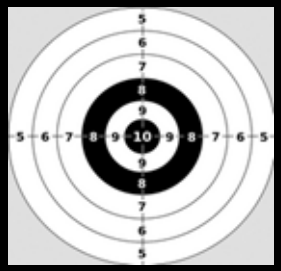
Barzan Mozafari is a Morris Wellman Assistant Professor of Computer Science and Engineering at the University of Michigan, Ann Arbor, where he leads a research group designing the next generation of scalable databases using advanced statistical models. Prior to that, he was a Postdoctoral Associate at MIT. He earned his Ph.D. in Computer Science from UCLA in 2011. His research career has led to many successful open-source projects, including CliffGuard (the first robust framework for database tuning), DBSeer (the first automated database diagnosis tool), and BlinkDB (the first massively parallel approximate query engine). He has won the National Science Foundation CAREER award, as well as several best paper awards in ACM SIGMOD and EuroSys. He is also a co-founder of DBSeer and a strategic advisor to SnappyData, a company that commercializes the ideas introduced by BlinkDB.

about projects publications vita bio students press coverage

My Research

Use ML to build smarter Data Systems:

More Efficient



More Predictable

More Manageable

FAIS Lab
(Fully Autonomous, Infinitely Scalable)

BARZAN MOZAFARI · UNIVERSITY OF MICHIGAN · DATABASE SYSTEMS

Current Projects

- **Infinite-Scale Analytics** : Reducing Computational Footprint with Approximate Query Processing
- **Designing a Predictable Database** : Reducing Tail Latencies in Modern Data-Intensive Applications
- **DBSeer** : Self-Driving Databases

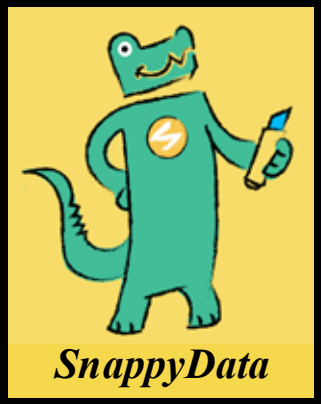
Research & Commercial Projects

keebo

AI agent for cloud databases

VerdictDB

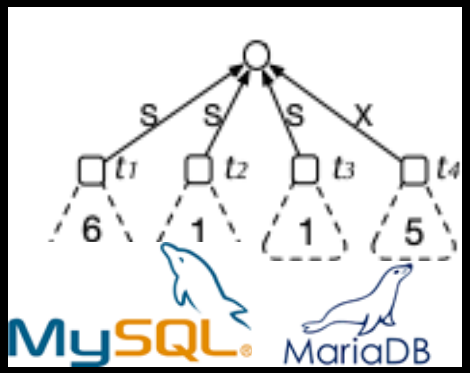
http://VerdictDB.org
Universal Approximate Query Processing



Stream, Transact, and Analyze in a Single Cluster

DBSeer

Using Machine Learning to Build Self-Driving Databases



Contention-Aware Lock Scheduling

BlinkDB

Querying Petabytes of Data in a Blink Time!

My Teaching

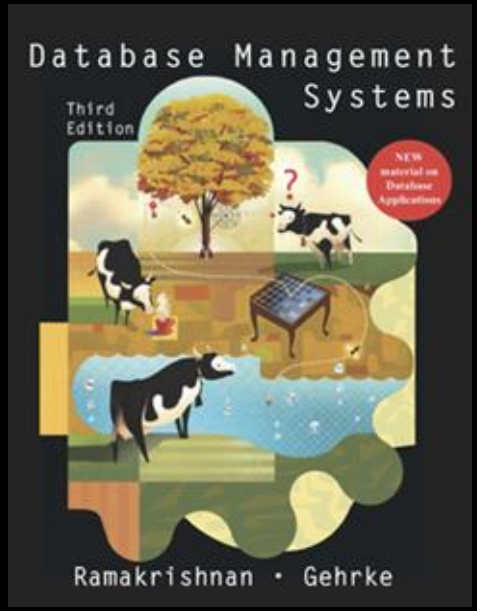
- Fun Fact: I've thought DBs at ALL levels! ☺
  - 484: Undergrad DB ~300 students
  - 584: Advanced DB ~30 students
  - 684: Super ultra crazy advanced DB ~3 students
- Why I Teach
  - Inspire students to go on and do great things in the world!
- Teaching Philosophy
  - You learn when you know why
  - You learn when you ask questions
  - You learn by doing

Course Overview

- Textbook: *The cow book*

  "Database Management Systems", by Raghu Ramakrishnan & Johannes Gehrke. 3rd ed.

- Prerequisites:
  - EECS 281
  - Data structures, e.g. hash tables, trees
  - Programming experience (C++, Java)



Instructional Staff: 3 GSI + 3 IAs

Name	Role
Samika Gupta	GSI
Jack Gong	GSI
Pritesh Murugesan Karthik	GSI
Colin Czarnik	IA
Allison Cui	IA
Yilin Chen	IA

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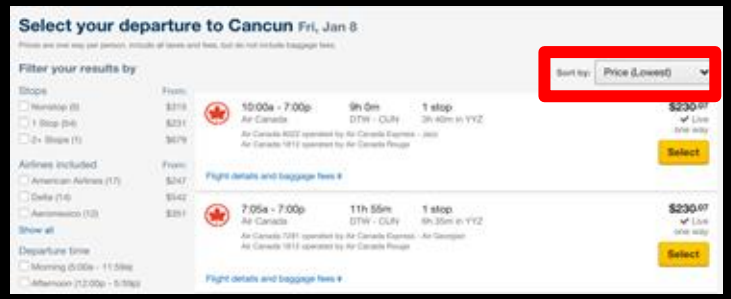
Why Study Databases?

- There are two types of people
  - Those who know SQL and those who don't
  - Data is useless without the tools to extract information (queries)
    - E.g.,"Optimal" pricing of flights



- Databases touch every aspect of our daily lives
  - Websites, social media, banks, digital libraries, online shopping, retail, mobile apps, games, human genome project, ...
- Databases touch most of CS
  - OS, languages, theory, AI, multimedia, logic, …

EECS 484: Course Overview

GOAL: Introduction to database management systems (DBMS)

- 1st half of semester:
  - Externals of DBMS (database user)

    How to use a DBMS, how to model data, how to store and retrieve info, how to search quickly for info, …
- 2nd half of semester:
  - Internals of DBMS (database implementer)

    Indexing, query optimization, transactions, …
  - Advanced Topics

    ???

DB User vs. DB Builder Perspectives

Databases are different than sausage

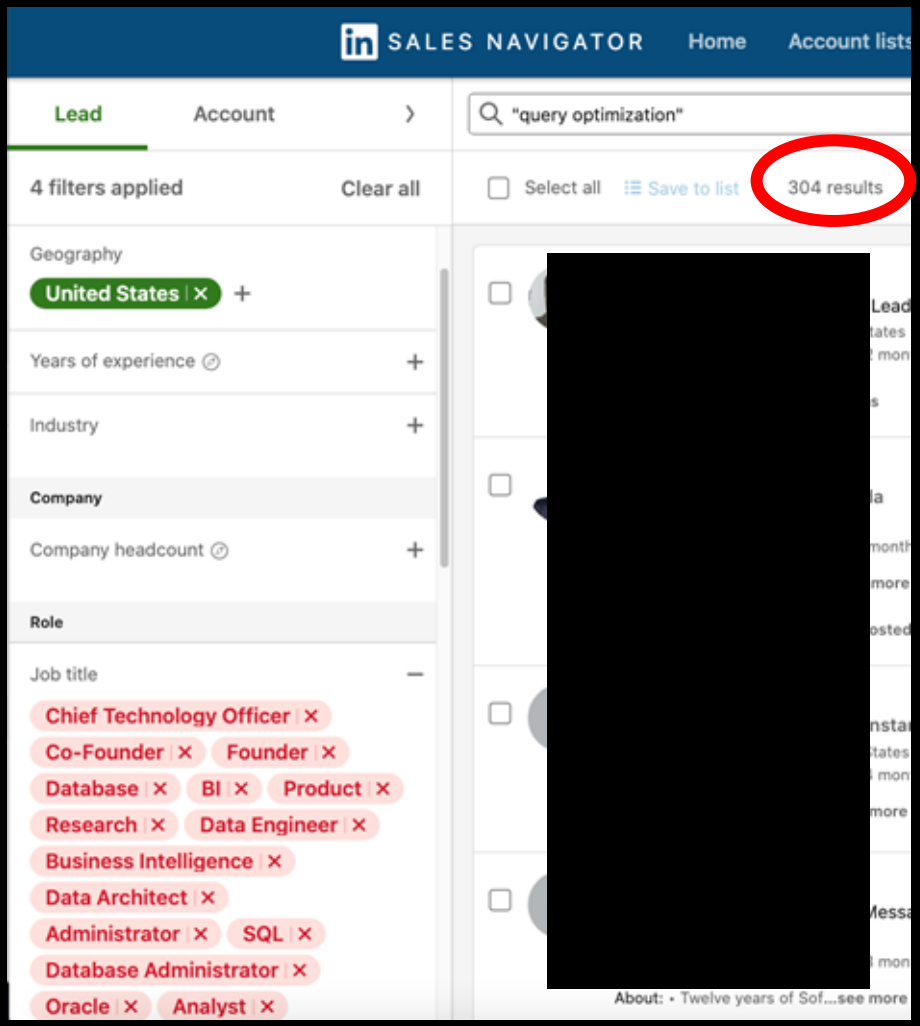
- You DO want to know how they're made!

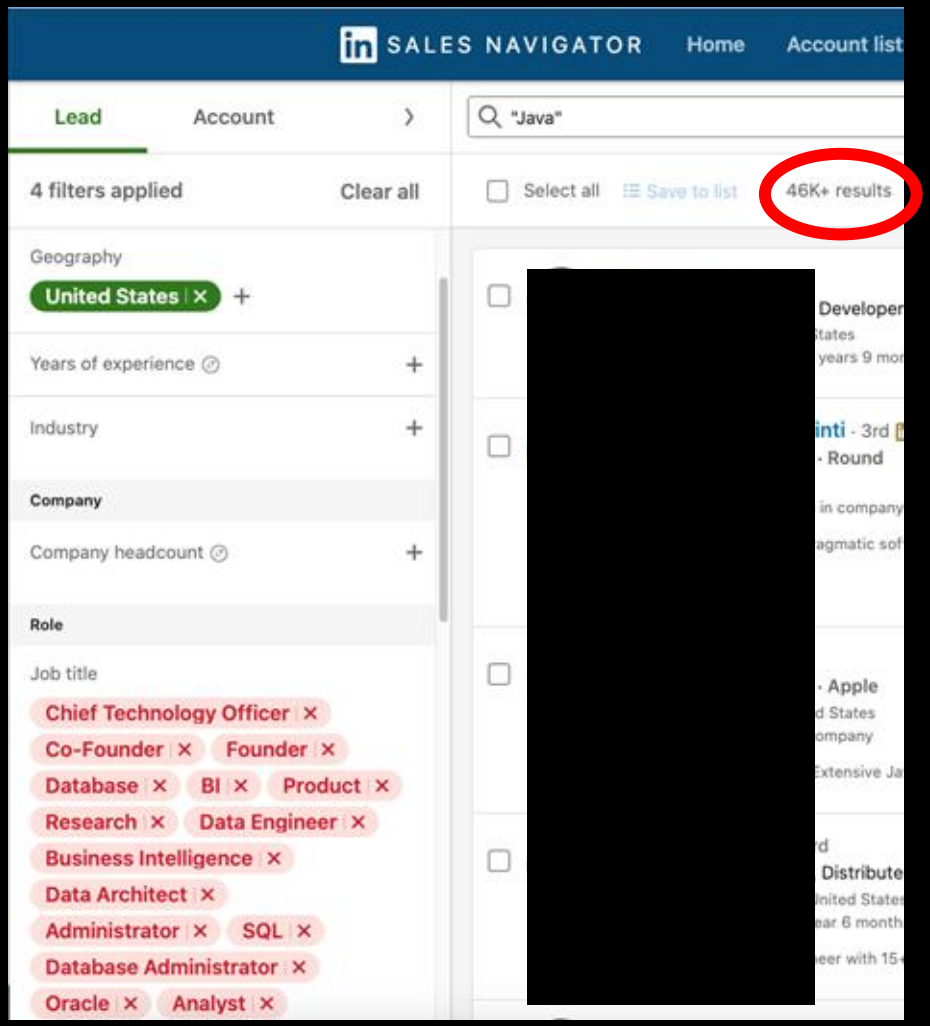
Why do we study DB internals?

1. Use DBs more effectively (know its limitations!)
2. Not all DBs are created equal
   - Pick the right hammer
3. Most complex breed of software systems
   - Learn about best system building skills
   - Perfect platform for learning many fundamental concepts

Golden Career Opportunities in DBs

Massive demand for engineers who understand DB internals and query optimizations!



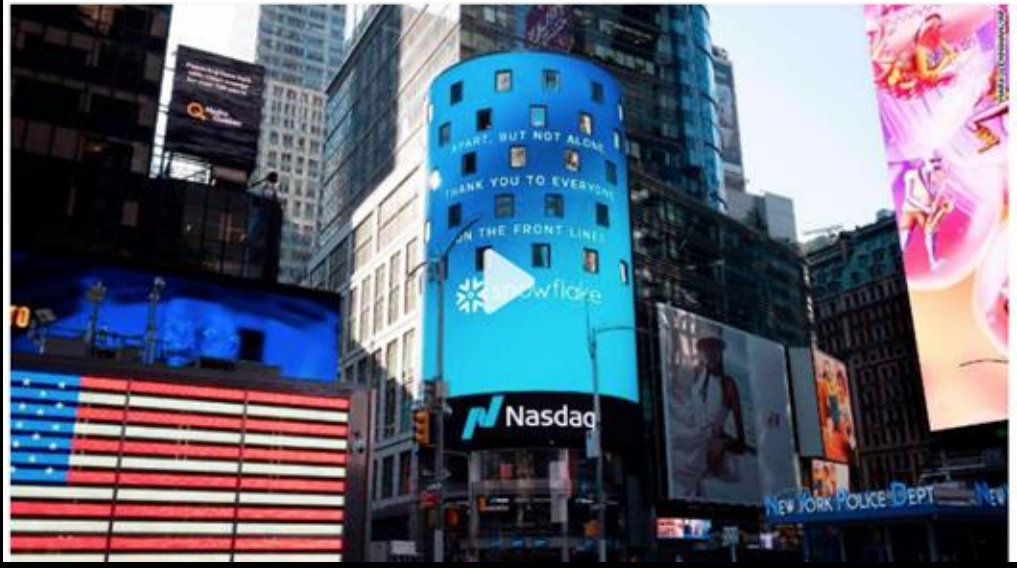


Massive, Growing Market

CNN BUSINESS · LIVE TV · CNN+

Snowflake shares more than double. It's the biggest software IPO ever

By Paul R. La Monica, CNN Business
Updated 3:37 AM ET, Thu September 17, 2020



Enterprise Database Market To Reach USD 155.50 Billion By 2026 | Reports And Data

Enterprise Database Market Size – USD 65.30 billion in 2018, Market Growth - CAGR of 11.1%, Industry Trends – Enhanced streamline business option.

July 23, 2019 10:32 ET | Source: Reports and Data

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Tentative Schedule

(subject to change)

Lec. # or Disc. #	Date	Topic	Readings	Assignments
L1	1/8	Introduction	Ch.1	
D0	1/9	No Discussion		
L2	1/13	ER Model	Ch.2	Project 1 release HW 1 release
L3	1/15	ER Model and Relational Model	Ch. 3.1–3.4 Ch. 5	
D1	1/16	ER Diagrams & Project 1 Intro		
L4	1/20	Intro to SQL Integrity Constraints	Ch. 5.1–5.2 Ch. 5.7	
L5	1/22	SQL DDL ER to Relational Triggers	Ch. 3.5 Ch. 5.8–5.9	
D2	1/23	SQL DDL & DML		
L6	1/27	ER to Relational Triggers	Ch. 4.1–4.2	HW 1 due HW 2 release
L7	1/29	Relational Algebra		
D3	1/30	More SQL		Project 1 due Mon. 2/2 Project 2 release Mon. 2/2
L8	2/3	Relational Algebra (cont.) SQL	Ch. 5	
L9	2/5	SQL Continued	Ch. 5 Ch 6.1–6.3, 6.5	
D4	2/6	Relational Algebra and P2 Intro	SQL Handout	
L10	2/10	DB App Programming; Schema Refinement and Normal Forms	Ch. 19.1–19.7	HW 2 due HW 3 release
L11	2/12	Schema Refinement and Normal Forms.	Ch. 8	
D5	2/13	Normalization and P2		
L12	2/17	Schema Refinement and Normal Forms.	Ch. 10.1–10.3 Ch. 10.4–10.8	
L13	2/19	Storage and Indexing		
D6	2/20	Midterm Review		Project 2 due Project 3 release
Exam 1	2/24	Midterm (First Half Exam), In person, 6-9 PM (No Lecture)		Details to be released on Piazza

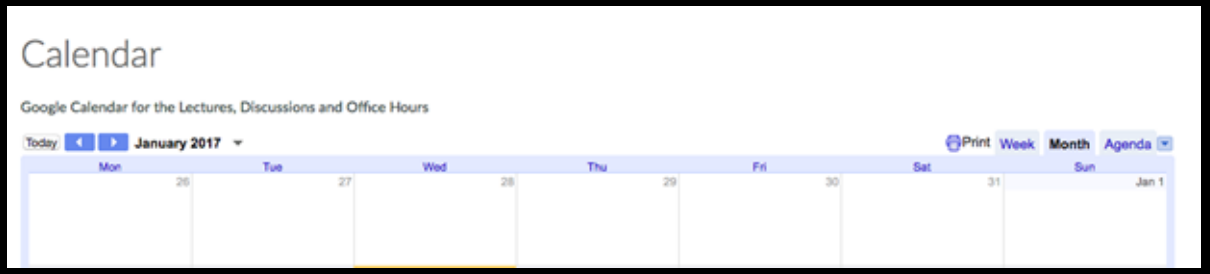
Tentative Schedule

(subject to change)

L14	2/26	Tree-structured indexing		
	2/27	No Discussion		
Spring Break	2/28-3/8	No Classes		
L15	3/10	Hash-based Indexing		
L16	3/12	External Sort	Ch. 11 & 13	
D7	3/13	P3 Intro + Background: NoSQL; MapReduce, Tree-Based Indexing		
L17	3/17	External Sort (cont.)	Ch 13	HW 3 due HW 4 release
L18	3/19	Query Evaluation: Selection	Ch. 12, 14	
D8	3/20	Indexing: Hash-based and Tree-based		
L19	3/24	Query Evaluation: Join	Ch. 12, 14	
L20	3/26	Query Evaluation: Join		
D9	3/27	Sorting		Project 3 due Project 4 release
L21	3/31	Query Evaluation: Other Operations Query Evaluation: Optimization	Ch. 15	HW 4 due HW 5 release
L22	4/2	Transaction Management Overview	Ch. 16	
D10	4/3	Joins; Project 4 Intro		
L23	4/7	Concurrency Control (zoom lecture)	Ch. 17 - 17.4	
L24	4/9	Logging and Recovery (zoom lecture)	Ch. 18	
D11	4/10	Optimization; Transaction		
L25	4/14	Logging and Recovery		HW 5 due
L26	4/15	Modern data intensive systems		
D12	4/16	Transaction; Logging and Recovery		Project 4 due
L27	4/21	Retrieval-Augmented Generation		
Exam 2	4/23	Final (Second Half Exam) In person, 7-9 PM, scheduled by the Registrar		You must take the exam at this time.

Office Hours: Calendar

- Check calendar before zooming to OHs



- Prof office hours
  - Proj- or HW-specific questions: go to TA OHs
    - E.g., “Why is my code not working?”
  - You can ask me anything else (material, concepts, general concerns, ...)

STRICT STRICT STRICT STRICT STRICT STRICT STRICT

Email Policy

STRICT STRICT STRICT STRICT STRICT STRICT STRICT

- Most questions (homework, projects): piazza
  - Check if it has been asked
  - Give answers (though, Honor Code)
    - Up to 2% bonus => karma points! ☺
  - Private piazza question ONLY if it would be violation of the Honor Code ---- NO CODE

STRICT STRICT STRICT STRICT STRICT STRICT STRICT

Email Policy

STRICT STRICT STRICT STRICT STRICT STRICT STRICT

- Special accommodations (personal questions)
  - 484questions@umich.edu
- Question for the prof or a specific TA?
  - 484questions@umich.edu
  - Any staff member can help you
  - Your question will be answered faster

Grading

No make up exams!

Midterm Exam	27%
Final Exam	27%
Projects: 4 programming assignments	40%
Homeworks: 5 written assignments	6%
Karma points: Bonus	0 to 2%
Karma points: Penalty	-5 to 0%

Grading

- The minimum you should expect:
  - The standard grading scale
  - But for C or higher, easonable performance on projects and exams is expected, irrespective of your overall %

Minimum percentage achieved	Grade
97%	A+
93%	A
90%	A-
87%	B+
83%	B
80%	B-
75%	C+
70%	C
65%	C-
55%	D+
45%	D
Below 45%	E

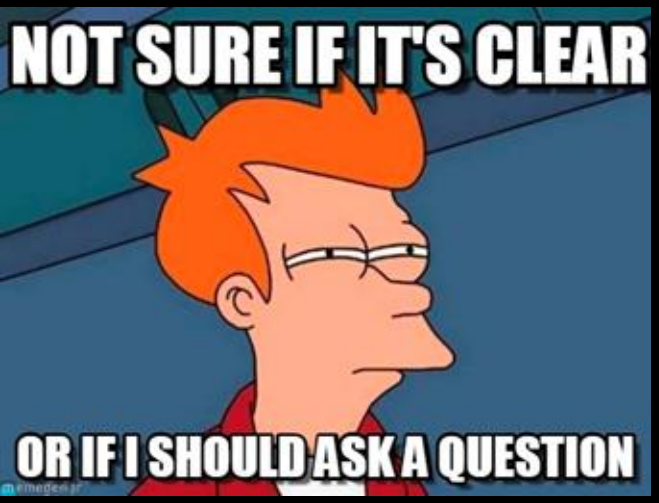
- Historical data
  - 25% get A- or higher, 50% get B's, the rest get C's & D's, ...

Exams

- Midterm Exam
  - Feb 24, 2025: 6-9pm (in-person)
- Final Exam
  - Apr 23, 2025: 7-9pm (in-person)

- Not cumulative (exception: tree-based hashing)

NO make up exams, plan your travel accordingly!



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Homeworks

- Total: 5 written assignments
- Done individually (no groups)

Projects

- Total: 4 coding projects

- Always in groups of two
- Start looking for partners now!
  use piazza
- Your partner can be in any of the two sections
- Can change partners from one proj to next

Late Days

- See course policy: https://docs.google.com/document/d/1dZaGicRUnmA7--zU2DIq5XLmJADskxjXqXQJTJyZuUA/edit?tab=t.0

- Submit early!

Project Repository

- To share the code with your partner:
  - Git – PRIVATE, otherwise violation of the Honor Code
  - Subversion
  - Google drive
  - Dropbox
  - Box
- Github: create and share with your partner a PRIVATE git repository, after requesting a student account. Alternative: bitbucket.
- Commit changes frequently!
- Version history helps document your contributions

Dos

- Lecture attendance is optional
  - BUT participation in class discussions or answering Piazza questions are both encouraged and REWARDED (up to +2% karma points)
- In-class quizzes meant to resemble your exams
  - And (at least) one question will appear on your exam verbatim!
- Take advantage of Piazza and Office Hours
  - Not just the day before the due date!

Don’ts

Rule 1. Do NOT violate your classmates’ learning rights

- side conversations or distracting behavior during lecture will NOT be tolerated since attendance is optional (-5% Karma points)

Rule 2. Do NOT violate your classmates’ grades

- Honor code violations will NOT be tolerated, honor council & stuff

Engineering Honor Code

- No reuse of old solutions from anywhere permitted
- Work on homework and projects must be entirely your own
- Only conceptual discussions allowed with others. No sharing of code.
- Help from TA/IA and instructors is OK
- No sharing of your solutions with others, even after the semester is over.
- No public github or other repositories
- Violations will be reported to the Engineering Honor Council
- We will use software and manual inspection to detect cheating on projects.

Lectures

- Attendance is optional
  - If in-person attendance exceeds 80% of room capacity, it will be live streaming to allow for virtual attendance
  - Otherwise, those who do not attend in person can watch recordings
- Slides will be uploaded **by 10:25am EST** on the day of class (see link on course schedule)
  - If needed, updated slides will be within 1 day
- Recording: Lectures are auto recorded
  - Don’t need to make your own recordings
  - Don’t share online
  - Lectures are copyrighted Material owned by U-M

Discussion Sections

- Not optional!
- Go with questions!
- Projects covered in discussion sections
- Exams may have project-related questions
- Multiple sections on Fridays:
  - identical content, but different instructors
- If you have to miss your section one week, attend another. But stick to your own as a rule

Project Topics

- Project 1: design and populate a database for Facebook data
  - We will provide (fake) data, but will give you the tools (and encourage you) to obtain your own data from your Facebook account
- Project 2: query this database, in a mix of SQL and Java
- Project 3: NoSQL (MongoDB) and querying documents with Javascript
- Project 4: implement Grace Hash Join in C++

Waiting List

- Attend the first 2 weeks of classes
- Even if you enroll late, homework and proj reqs are the same
  - No waiver or deadline extensions for late enrollment

- EECS advising office handles ALL requests for permissions / overrides
  - ugadmin@eecs.umich.edu
  - [I think] they usually give priority to senior students in CS-Eng/LSA who need the course to graduate this semester

Other Logistics

- Frequent questions and in-class quizzes
  - Think, Pair, Share
- Student questions are **ALWAYS** welcome
  - Help me help you!
- My 90% Rule
  - I move on when 90% of class are clear about 90% of the material
- One 5-min bio break per lecture
  - Settle back in **quickly** at the end of the 5 mins!

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